

A Game Theoretic Approach to Phase Balancing by Plug-In Electric Vehicles in the Smart Grid

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Abstract—With the increasing penetration of renewable energy resources and plug-in electric vehicles (PEV) in the smart grid, the problem of phase imbalance in a three-phase distribution system becomes more challenging. In this paper, we propose a phase balancing (PB) scheme performed by PEVs. With PB remunerations introduced as financial incentives, PEV owners are encouraged to assist the distribution system operator in reducing the phase imbalance. This is achieved by exploiting the charging flexibility of PEV batteries. We formulate this problem into a non-smooth non-cooperative game, where each PEV owner autonomously minimizes his/her charging cost minus the PB remuneration, considering decisions of other PEV owners. Through transforming this game into a smooth extended one, we are able to solve this challenging problem. Specifically, existence of its Nash equilibrium (NE) is demonstrated, and a distributed algorithm is developed to achieve the NE. Convergence proof of this algorithm is also provided. Simulation results validate our proposed scheme, and illustrate that both PEV owners and the distribution system operator will benefit financially. Moreover, the power quality and system reliability of the distribution network can also be improved.

Index Terms—Phase balancing, plug-in electric vehicles, game theory, variational inequality, distributed algorithm.

I. INTRODUCTION

DUE TO the large number of unequal single-phase loads, three-phase distribution system is inherently unbalanced [1]. Load imbalance among different phases can bring adverse impacts on the power system and electrical equipments therein. Under unbalanced conditions, currents in some phases

as well as in the neutral line will increase, thus incurring more energy losses and heating effects [2], even leading to tripping of the distribution feeder and overload-protection circuits [3]. Phase imbalance can also reduce the available capacity of feeders and transformers, causing additional investment costs to the power system [4]. Furthermore, an increase in the phase imbalance will deteriorate the power quality, decrease efficiency and lifetime of the induction motor, and generate non-characteristic harmonics from electronic devices [5].

In recent years, the penetration of distributed renewable energy generators (REGs), e.g. rooftop Photovoltaic (PV) panels and residential wind turbines, is growing around the world. Since the power generation of REG is intermittent and uncertain, and REGs installed by electricity customers are randomly distributed among the phases, the imbalance problem is becoming more severe [6], [7]. Therefore, the distribution system operator (DSO) should take measures to balance three phases dynamically, in order to improve the power quality and system reliability of the distribution network.

A lot of research efforts have been made to solve the PB problem. Traditionally, rephasing techniques which re-assign loads to different phases are widely used [8]. Indeed, [8] introduced a mixed-integer programming formulation to select the optimal rephasing decision. This framework was extended in [9] to account for the presence of demand-side uncertainty. In [10], optimal rephasing strategies were determined by solving a multi-objective optimization problem considering rephasing cost and voltage drop. Furthermore, Gray *et.al.* investigated the economic feasibility of the rephasing approach in [11], where a particle swarm optimization problem was solved to mitigate the power imbalance.

These papers mentioned above all considered manual rephasing operations, which will incur significant costs due to the required manual labor and customer service interruption. As a result, these research works are insufficient to address the dynamic PB challenges in the future smart grid. Consequently, static transfer switches (STS) were introduced in [12] to dynamically transfer residential loads of each house from one phase to another. Reference [13] also performed dynamic PB through the installation of automatic re-phasing switches. In [14], automatic rephasing and distributed generation sizing were jointly investigated. However, these novel rephasing approaches involve additional investments during their implementation. What is more, according to [15], three-phase power imbalance can be decomposed into two components: systematic imbalance and

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random imbalance. Rephasing techniques can only address the first part, thus other PB approaches need to be pursued to handle the remaining one.

Recently, many studies have been performed to mitigate power imbalance based on demand-side resources (DSR). The key idea is to exploit the power generation and consumption flexibility of DSR to balance loads. In [16], three-phase PV inverters and PEV chargers were adapted to transfer power from highly loaded phases to less loaded ones. Sun *et.al.* [17] effectively solved the PB problem by scheduling the charging and discharging of energy storages. In [18], the authors proposed an optimization-based paradigm to reduce the phase imbalance by controlling the operation of data centers. Coordination of distributed single-phase inverters was performed in [19] so that the surplus power capability of DSR is utilized to compensate unbalanced phases. Single-phase inverters were also used in [20] to mitigate imbalance by modifying the reactive power production. Reference [21] proposed two platforms that connect DC sides of adjacent PV inverters and PEV chargers in different phases together, such that loads can be transferred among the phases. In [22], PV inverters and thermostatically controlled loads were jointly coordinated to mitigate the phase imbalance.

In general, DSRs are possessed by electricity customers rather than the DSO. Therefore, appropriate incentive schemes should be designed to encourage the participation of customers. Otherwise, implementation of PB frameworks discussed above will be difficult. However, none of these papers has considered this critical issue. Different from existing works, we investigate the PB problem from the perspective of PEV owners, where incentives are provided to encourage their participation in PB. Then we formulate the PB problem as a game of self-interested PEV owners. Note that we choose a game-theoretic setting rather than an optimization-based one to capture the autonomy and rationality of PEV owners. Due to the non-smooth terms introduced by PB incentives, this game is very challenging to tackle. Hence, we manage to analyze its properties by studying an extended game instead. Our main contributions are summarized as follows:

- We introduce PB incentives to the PEV owners and model their interactions as a non-cooperative PB game.
- An extended game is proposed to circumvent the significant challenge of handling the original PB game. Connections between these two games are built, and existence of their Nash equilibrium (NE) is proved.
- A distributed algorithm is developed to achieve the NE, and its convergence proof is also provided.
- Benefits of our proposed scheme to both PEV owners and the DSO are illustrated through extensive simulations.

The remainder of this paper is organized as follows. Section II presents the system model and formulates the PB problem as a game. In Section III, a smooth extended game is introduced to assist in analyzing the original game and developing a distributed algorithm to achieve its NE. Numerical results are presented in Section IV and Section V concludes this paper.

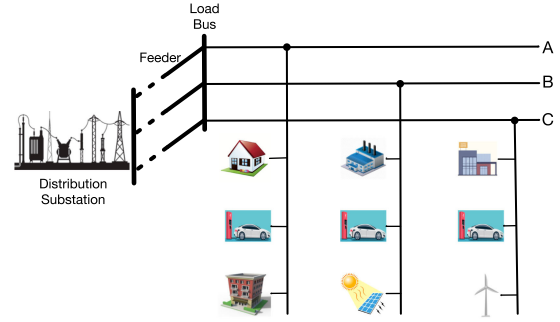


Fig. 1. A three-phase distribution system. Single-phase loads and REGs in a neighborhood are connected to the same load bus, while this bus is connected to other buses and the substation through three-phase distribution feeders.

II. SYSTEM MODEL

In a three-phase distribution system with distributed renewable energy and PEV penetration as shown in Fig. 1, single-phase loads and REGs such as PV panels and wind turbines are connected to different phases of a load bus. Load buses are connected to the substation through three-phase distribution feeders. Due to the variability and uncertainty of REG production, the power loads on three phases can be highly unbalanced.

Assume there are a set of PEV owners $\mathcal{N} = \{1, 2, \dots, N\}$ who voluntarily participate in the PB scheme. We first suppose these PEVs charge their batteries in a same small neighborhood. All the loads and REGs in this neighborhood are connected to the same load bus. Compared with the power flows in the feeders of the larger distribution system, power flows inside this small neighborhood have minor effects on the load voltages. Moreover, under normal conditions, electrical lines in the neighborhood usually have enough capacity to hold all the loads. Therefore, we do not consider the power flow constraints within this load bus. However, we will discuss later how our proposed PB scheme can be integrated into the larger distribution system where power flow constraints should not be ignored.

Each PEV $i \in \mathcal{N}$ is connected to one phase.¹

Clearly, the set $\mathcal{N}$ consists of three subsets, i.e. $\mathcal{N} = \bigcup_{j \in \Phi} \mathcal{N}_j$, where $\Phi = \{1, 2, 3\}$ and each subset $\mathcal{N}_j$ represents the charging PEVs connected to a particular phase j . Note that we use phase 1, 2, 3 to denote phase A, B, C respectively. Since many PEVs, especially those used for commuting, spend most of the time parking at home or the workplace, their charging time can be rather flexible. Indeed, the key idea of PEV-based PB is to exploit the time flexibility of PEV charging. To be specific, the DSO provides certain incentives to encourage the rescheduling of PEV charging, such that charging loads on each phase are shifted to the time-slots when this phase is relatively light-loaded compared with the other two.

¹Due to their convenience and low infrastructure costs [23], single-phase chargers are popular for homes and business sites. As for three-phase DC fast chargers, when the charging power on each phase is controllable, as proposed in [16], it can be regarded as three single-phase chargers with a common DC side. With the energy demands of PEVs extracted from each phase pre-determined, our proposed scheme can be extended to the case where three-phase chargers are used.

Suppose that a PEV $i \in \mathcal{N}$ needs to charge its battery to a certain state-of-charge (SOC) D_i over a time horizon $\mathcal{T} = \{1, 2, \dots, T\}$, where $0 \leq D_i \leq 1$, and $t \in \mathcal{T}$ is a time-slot, e.g., 15 mins. Denote the charging rate of PEV i in time-slot t as x_t^i , the following constraint must be satisfied:

$$D_i B_i \leq S_i^0 B_i + \sum_{t=1}^T x_t^i \leq B_i, \quad (1)$$

where B_i is the battery capacity of PEV i , and S_i^0 is the initial battery SOC. The charging rate is also constrained:

$$0 \leq x_t^i \leq x_{\max}^i, \quad (2)$$

where $x_{\max}^i$ is PEV i 's maximum charging rate. Therefore, the feasible charging rates $\mathbf{x}^i = (x_1^i, \dots, x_T^i)^T$ of PEV i can be described by a set Ω_i :

$$\Omega_i = \left\{ \mathbf{x}^i \in \mathbb{R}^T : D_i B_i \leq S_i^0 B_i + \sum_{t=1}^T x_t^i \leq B_i, \right. \\ \left. 0 \leq x_t^i \leq x_{\max}^i, \forall t \in \mathcal{T} \right\}. \quad (3)$$

As Ω_i is the solution set of a finite number of linear inequalities, by definition, it is a polyhedron which is convex [24]. Since it is both closed and bounded, we can conclude that Ω_i is also compact [25].

As REG can be regarded as negative loads, we define the inflexible load $\tilde{L}_t^j$ on phase j in time-slot t as the conventional power load less the REG power generation.

Note that the conventional power load here excludes the flexible PEV loads committed to PB.

Hence, the total inflexible load $\tilde{L}_t$ is the sum of inflexible loads on three phases, i.e., $\tilde{L}_t = \sum_{j \in \Phi} \tilde{L}_t^j$. Similarly, the total load L_t is defined as $L_t = \tilde{L}_t + \sum_{i \in \mathcal{N}} x_t^i$, i.e., the sum of inflexible loads and PEV loads. Here we first assume the inflexible loads are known or can be predicted accurately, i.e., $\tilde{L}_t^j$ is a deterministic parameter. Later we will discuss how to tackle the load uncertainty in practice.

We assume the electricity price $p_t(L_t)$ in time-slot t is a non-decreasing twice differentiable convex function of L_t . Equivalent pricing models are given in [26], [27]. Under this pricing scheme, the corresponding electricity cost function is a generalization of the widely used linear and quadratic electricity cost functions in the power system literature. Therefore, the charging cost of each PEV i over $\mathcal{T}$ is $\sum_{t=1}^T p_t(L_t) x_t^i$.

In order to encourage PEV owners' participation in phase balancing, financial incentives must be provided to them. Similar with the definitions in [9] and [17], we define the phase imbalance UB_t as

$$UB_t = \max_{j \in \Phi} \left\{ (L_t^j - L_t/3)^2 \right\}, \quad (4)$$

where $L_t/3$ is the average total load of each phase. Accordingly the original imbalance without committed PEV loads is

$$UB_t^o = \max_{j \in \Phi} \left\{ (\tilde{L}_t^j - \tilde{L}_t/3)^2 \right\}. \quad (5)$$

The gross remuneration R_t given to the PEV owners for PB is assumed to be proportional to the imbalance reduction,² i.e.,

$$R_t = \kappa_t (UB_t^o - UB_t), \quad (6)$$

where κ_t is a pricing coefficient in time-slot t . As the individual contribution of each PEV to phase imbalance reduction is difficult to measure, for simplicity in implementation, we assume that each PEV owner acquires an equal share of the gross remuneration, which is R_t/N . Note that PB participation is voluntary rather than compulsory, PEV owners can decide if they will join the scheme and accept this equal share. Naturally, the objective of each PEV owner i in the PB scheme is to minimize the charging cost $\sum_{t=1}^T p_t(L_t) x_t^i$ and maximize the phase balancing remuneration $\sum_{t=1}^T R_t/N$ by scheduling the charging rate x_t^i . As inflexible loads are known parameters, UB_t^o is a constant, thus we can drop it. To summarize, PEV owner i solves the following problem:

$$\min_{\mathbf{x}^i \in \Omega_i} \sum_{t=1}^T (p_t(L_t) x_t^i + \kappa_t \cdot UB_t/N). \quad (7)$$

From (7), we can see that κ controls the preference of PEV owners for participating in PB. When κ is large, PEV owners are more willing to help in minimizing the phase imbalance.

Let

$\mathbf{x}_t^{-i} = (x_t^1, \dots, x_t^{i-1}, x_t^{i+1}, \dots, x_t^N)^T \in \mathbb{R}^{N-1}$ denote the collective charging rates of PEVs excluding PEV i . To simplify the writing, we introduce a new function g as follows:

$$g_t^j(x_t^i, \mathbf{x}_t^{-i}) = \left(\tilde{L}_t^j + \sum_{i \in \mathcal{N}_j} x_t^i - L_t/3 \right)^2,$$

then we have $UB_t = \max_{j \in \Phi} g_t^j$, and (7) can be rewritten into the following problem:

$$\min_{\mathbf{x}^i \in \Omega_i} \sum_{t=1}^T \left(p_t(x_t^i, \mathbf{x}_t^{-i}) x_t^i + \frac{\kappa_t}{N} \max_{j \in \Phi} g_t^j(x_t^i, \mathbf{x}_t^{-i}) \right). \quad (8)$$

Obviously, when all PEV owners try to solve problem (8), we can formulate this phase balancing process as a Nash game, whose formal definition is as follows:

Phase Balancing (PB) Game Γ :

- **Players:** N PEVs.
- **Strategies:** PEV owner i determines the optimal charging rates $\mathbf{x}^i$ given the other PEV owners' decisions $\mathbf{x}^{-i}$ by solving problem (8).

Remark: Game theoretic methods have the capability to model strategic behaviors of the players. As PEV owners are rational, they will make their own decisions to maximize their welfare considering all the information available, including other owners' behaviors. Under this circumstance, it is straightforward to model this PB problem as a game. Instead, if considering the PB problem from the perspective of the DSO, a centralized optimization-based approach can be used. However, strategic behaviors of the PEV owners are ignored, which may not be

²We select the linear function for its technical and practical simplicity. More complicated functions can be too difficult to implement in reality.

realistic in practice. The reason is that PEV owners lack the incentives to follow the directions of the DSO.

Since the finite minmax term in (8) renders the objective function non-differentiable, the state-of-the-art distributed algorithms [28], [29] for standard Nash games are not applicable. Following the techniques in Section 4.3.1 of [24], we can move this term from the objective function to the constraints by forming a corresponding epigraph problem. However, the resulting problem will become a generalized Nash game, which is still exceptionally difficult to solve [30].

III. PHASE BALANCING GAME

As previously explained, this PB game is difficult to address because of the nonsmooth minmax terms. In this section, we first reformulate it into a smooth game, then establish its NE existence result through this smooth version, and finally a distributed algorithm is developed for NE computation.

A. Reformulation of the PB Game

Let us consider problem (8), it is easy to see that

$$\max_{j \in \Phi} g_t^j(x_t^i, \mathbf{x}_t^{-i}) = \max_{\mathbf{y}_t \in \Delta_t} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}), \quad (9)$$

where $\mathbf{y}_t = (y_t^1, y_t^2, y_t^3)^T \in \mathbb{R}^3$, and $\Delta_t = \{\mathbf{y}_t : \sum_{j \in \Phi} y_t^j = 1, \mathbf{y}_t \geq 0\}$. As with Ω_i , the set Δ_t is also a polyhedron that is both convex and compact. Hence, the following optimization problem is equivalent to problem (8):

$$\min_{\mathbf{x}^i \in \Omega_i} \sum_{t=1}^T \left(p_t(x_t^i, \mathbf{x}_t^{-i}) x_t^i + \frac{\kappa_t}{N} \max_{\mathbf{y}_t \in \Delta_t} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}) \right). \quad (10)$$

In order to solve problem (10), we introduce an extended Nash game $\hat{\Gamma}$ by constructing T artificial players (AP) to decompose the nonsmooth minmax terms away from the optimization problems of PEV owners. In practice, these APs' roles can be performed by the DSO. $\hat{\Gamma}$ is presented as follows:

Extended PB Game $\hat{\Gamma}$:

- *Players:* N PEVs, T APs.
- *Strategies:* PEV i determines the optimal charging rates $\mathbf{x}^i$ given the other PEVs' decisions $\mathbf{x}^{-i}$ and APs' decisions $\mathbf{y}_t, t \in \mathcal{T}$ by solving the following problem:

$$\min_{\mathbf{x}^i \in \Omega_i} \sum_{t=1}^T \left(p_t(x_t^i, \mathbf{x}_t^{-i}) x_t^i + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}) \right), \quad (11)$$

AP t determines the optimal $\mathbf{y}_t$ given PEVs' charging rates $\mathbf{x}^i, i \in \mathcal{N}$ by solving the following problem:

$$\min_{\mathbf{y}_t \in \Delta_t} - \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}). \quad (12)$$

Next we establish the connection between Γ and $\hat{\Gamma}$ in terms of their equilibria:

Theorem 1: If $(\mathbf{x}^{*,1}, \dots, \mathbf{x}^{*,N}, \mathbf{y}_1^*, \dots, \mathbf{y}_T^*)$ is an NE of the game $\hat{\Gamma}$, then $(\mathbf{x}^{*,1}, \dots, \mathbf{x}^{*,N})$ is an NE of the game Γ .

Proof: Please refer to Appendix A. ■

This theorem offers us a way to investigate the NE of game Γ by studying a smooth game $\hat{\Gamma}$ which is easier to solve.

B. Variational Inequality Based Equilibrium Analysis

Based on the previous subsection, we can analyze the game $\hat{\Gamma}$ instead of the original PB game Γ .

Denote the objective function of PEV i and AP t in game $\hat{\Gamma}$ as $H_i(\mathbf{x}^i, \mathbf{x}^{-i}, \mathbf{y})$ and $I_t(\mathbf{y}_t, \mathbf{y}_{-t}, \mathbf{x})$ respectively,

$$H_i(\mathbf{x}^i, \mathbf{x}^{-i}, \mathbf{y}) = \sum_{t=1}^T \left(p_t(x_t^i, \mathbf{x}_t^{-i}) x_t^i + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}) \right), \quad (13)$$

$$I_t(\mathbf{y}_t, \mathbf{y}_{-t}, \mathbf{x}) = - \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}), \quad (14)$$

where $\mathbf{x} \in \mathbb{R}^{NT}$ and $\mathbf{y} \in \mathbb{R}^{3T}$ are the collective strategies of PEVs and APs respectively. From the convexity and differentiability assumptions of the pricing function p_t in Section II, we can derive the properties of $H_i(\mathbf{x}^i, \mathbf{x}^{-i}, \mathbf{y})$ and $I_t(\mathbf{y}_t, \mathbf{y}_{-t}, \mathbf{x})$ in the following lemma:

Lemma 1: For each fixed $\mathbf{x}^{-i}$ and $\mathbf{y}$, $H_i(\mathbf{x}^i, \mathbf{x}^{-i}, \mathbf{y})$ is a convex and continuously differentiable function of $\mathbf{x}^i$. For each fixed $\mathbf{y}_{-t}$ and $\mathbf{x}$, $I_t(\mathbf{y}_t, \mathbf{y}_{-t}, \mathbf{x})$ is a convex and continuously differentiable function of $\mathbf{y}_t$.

Proof: Please refer to Appendix B. ■

Moreover, we have already known that both Ω_i and Δ_t are convex and compact. Therefore, based on Proposition 1.4.2 in [31], the extended PB game $\hat{\Gamma}$ can be written as a variational inequality VI($\mathbf{K}, \mathbf{F}$). In this VI, $\mathbf{K} = \prod_{i=1}^N \Omega_i \times \prod_{t=1}^T \Delta_t$, and

$$\mathbf{z} \equiv (\mathbf{x}^1, \dots, \mathbf{x}^N, \mathbf{y}_1, \dots, \mathbf{y}_T)^T \in \mathbf{K},$$

$$\mathbf{F}(\mathbf{z}) \equiv (\nabla_{\mathbf{x}^1} H_1, \dots, \nabla_{\mathbf{x}^N} H_N, \nabla_{\mathbf{y}_1} I_1, \dots, \nabla_{\mathbf{y}_T} I_T)^T.$$

A tuple $\mathbf{z} \in \mathbf{K}$ is an NE of $\hat{\Gamma}$ if and only if $\mathbf{z}$ belongs to the solution set of VI($\mathbf{K}, \mathbf{F}$), i.e. $\mathbf{z} \in \text{SOL}(\mathbf{K}, \mathbf{F})$.

Building on the VI formulation of $\hat{\Gamma}$ and Theorem 1, we are able to state the following corollary:

Corollary 1: Game Γ possesses at least one NE.

Proof: By Corollary 2.2.5 in [31], the set $\text{SOL}(\mathbf{K}, \mathbf{F})$ is nonempty, hence the game $\hat{\Gamma}$ has at least one NE. The final result readily follows from Theorem 1. ■

C. A Distributed Algorithm for the PB Game

Basically, in game-theoretic regimes, players should be able to independently make their own decisions, rather than being enforced by a central controller. As a result, to compute the NE, we will pursue a distributed algorithm. Since existing distributed algorithms for solving VIs generally rely on the monotonicity property of the mapping $\mathbf{F}$, we first present a sufficient condition for $\mathbf{F}$ to be monotone:

Lemma 2: $\mathbf{F}(\mathbf{z})$ is monotone when $p'_t \geq NX_{\max} p''_t, \forall t \in \mathcal{T}$, where p'_t and p''_t are first-order and second-order differentials respectively, while $X_{\max} = \max_{i \in \mathcal{N}} x_{\max}^i$ is the maximum charging rate.

Proof: Please refer to Appendix C. ■

Algorithm 1: Distributed Best-response Algorithm.

-
- 1: Let ϵ, δ, η and $\tau > 0$ be given.
 - 2: Set the counter $n = 1$. Set the initial $\hat{\mathbf{x}}_n = \mathbf{0}$ and $\hat{\mathbf{y}}_n = \mathbf{0}$.
 - 3: **repeat**
 - 4: **repeat**
 - 5: Set the counter $k = 1$.
 - 6: **for** $i = 1$ to N **do**
 - 7: PEV i solves problem (15) with $\hat{\mathbf{x}}^i = \mathbf{x}_n^i$, and sends the optimal solution $\mathbf{x}_{k+1}^i$ to the DSO.
 - 8: **end for**
 - 9: The DSO solves each AP t 's problem (16) with $\hat{\mathbf{y}}_t = \mathbf{y}_{t,n}$ and sends the optimal solution $\mathbf{y}_{t,k+1}$ together with $\mathbf{x}_{k+1}^i$ to each PEV i .
 - 10: $k = k + 1$.
 - 11: **until** $\|\mathbf{x}_k - \mathbf{x}_{k-1}\| < \delta$ & $\|\mathbf{y}_k - \mathbf{y}_{k-1}\| < \delta$.
 - 12: PEV i updates $\mathbf{x}_{n+1}^i = (1 - \eta)\mathbf{x}_n^i + \eta\mathbf{x}_k^i$, and the DSO updates $\mathbf{y}_{t,n+1} = (1 - \eta)\mathbf{y}_{t,n} + \eta\mathbf{y}_{t,k}$ for each t .
 - 13: $n = n + 1$.
 - 14: **until** $\|\mathbf{x}_n - \mathbf{x}_{n-1}\| < \epsilon$ & $\|\mathbf{y}_n - \mathbf{y}_{n-1}\| < \epsilon$.
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Remark: This sufficient condition is satisfied by many widely-used pricing functions, e.g., the constant and linear pricing function. Hence, it is not difficult to fulfill in practice.

With $\mathbf{F}(\mathbf{z})$ being monotone, we can develop a distributed best-response algorithm based on the regularization technique proposed in [29]. The basic idea is to solve a sequence of regularized strongly monotone VIs instead of a single monotone VI. To be specific, in each regularized VI, PEV i solves the following problem:

$$\min_{\mathbf{x}^i \in \Omega_i} \sum_{t=1}^T \left(p_t(x_t^i, \mathbf{x}_t^{-i})x_t^i + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}) \right) + \frac{\tau}{2} \|\mathbf{x}^i - \hat{\mathbf{x}}^i\|^2, \quad (15)$$

where $\hat{\mathbf{x}}^i \in \mathbf{R}^T$ is a given vector, while AP t solves:

$$\min_{\mathbf{y}_t \in \Delta_t} -\frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{-i}) + \frac{\tau}{2} \|\mathbf{y}_t - \hat{\mathbf{y}}_t\|^2, \quad (16)$$

where $\hat{\mathbf{y}}_t \in \mathbf{R}^3$ is also a given vector. In both formulations, τ is defined as a large constant whose role will be elaborated later. Details of this algorithm are presented in Algorithm 1, where both ϵ and δ are very small constants that act as termination criteria.

Next we define two constants as follows:

$$P_1 = \max_{t \in \mathcal{T}, l \in [0, L]} p'_t(l), \quad P_2 = \max_{t \in \mathcal{T}, l \in [0, L]} p''_t(l),$$

where L is the maximum allowable load in this system.³ The following theorem presents the convergence result:

³In practice, the electricity price will not increase in an infinite large rate, therefore P_1 and P_2 are well-defined finite constants.

Theorem 2: When $p'_t \geq NX_{\max} p''_t, \forall t \in \mathcal{T}$ holds, if the constant η satisfies $0 < \eta < 2$, and the regularization coefficient τ satisfies

$$\tau > N^{\frac{3}{2}} (P_1 + P_2 X_{\max}) + N\kappa_M + \frac{4T\kappa_M}{3N} \sqrt{\frac{2}{3}} (NX_{\max} + \tilde{L}_{\max}),$$

then Algorithm 1 converges to an NE of Γ , where κ_M is the maximum pricing coefficient, $\tilde{L}_{\max}$ is the maximum inflexible load.

Proof: Please refer to Appendix D. ■

Remark: When Algorithm 1 converges, the regularization term in (15) is negligible due to the termination criterion. Then, it can be ignored and (15) becomes the objective function solved by each PEV owner as shown in (13). Therefore, the result obtained by Algorithm 1 conforms with the NE of the PB game.

D. Discussions

We have already known in Section II that parameter κ controls the preference of PEV owners for participating in PB. When κ is a large number, PEV owners are more motivated to help the DSO reduce the phase imbalance, while they care more about minimizing their own energy cost when the value of κ is small. Therefore, by changing the value of κ at a load bus, the DSO can control the PEV charging behaviors at this bus for its benefit. Roughly speaking, the DSO may comply with the following heuristic: increasing the value of κ leads to more phase imbalance reduction.

In order to implement our proposed PB scheme, a two-level hierarchical structure can be introduced. In the upper level, the DSO determines the appropriate values of κ and publishes them to the PEV owners. First, it collects the current and predicted future information of loads and REGs at each bus. In this way, it acquires the potential power injection of each phase at all the buses. Combining the above heuristic of κ and experiences from historical implementations, the DSO chooses a large κ for the highly unbalanced bus and a small κ for the relatively balanced bus.⁴ In the lower level, the proposed PB game is performed among PEV owners at each bus with the published value of κ from the upper level. After that the charging rates of each PEV are decided.

In practice, the future load information is unknown and the prediction may not be accurate. To tackle this load uncertainty, we can use a receding horizon based strategy as follows: At each time-slot, the DSO announces the PB incentive, i.e., the value of κ for the considered time horizon. Then the PB game is performed with the known load information of the current slot and the inaccurate information of future slots, such as mean values of the load power. Only the output of the current time

⁴Ideally, κ should be determined exactly by solving a two-level problem, where the lower level is the PB game, while the upper level is an OPF problem. The DSO solves the OPF problem to obtain the optimal values of κ with the equilibria set of the PB game being a constraint. In this case, this problem becomes a mathematical program with equilibrium constraints (MPEC) which is even more challenging than the current PB game [32]. This problem is beyond the scope of this paper, and we have left it as our future work.

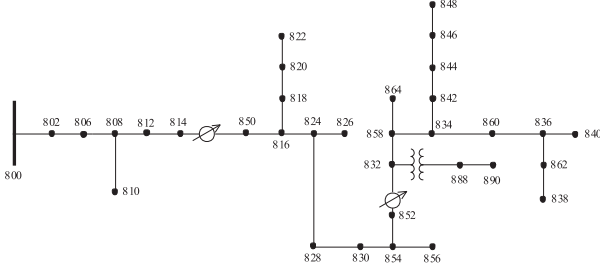


Fig. 2. IEEE 34-bus test feeder.

TABLE I
MEAN INFLEXIBLE LOADS AT BUS 846 (UNIT: kW)

Time-slot	Phase A	Phase B	Phase C
1-4	150	50	250
5-8	150	250	50

slot is applied. At the next time-slot, the DSO can update the PB incentive, and the game is performed again with a shifted time horizon.

IV. NUMERICAL RESULTS

In this section, we numerically evaluate the performance of our proposed PB scheme. We consider a period of 2 hours with each time-slot being 15 minutes, then the time horizon is $\mathcal{T} = \{1, 2, \dots, 8\}$. The IEEE 34-bus test feeder [33] shown in Fig. 2 is modified to serve as an example. This network is a four-wire three-phase feeder with isolated neutral (without ground conductor), which is the same with test case A studied in [34]. When the neutral line is multi-grounded, a portion of the neutral current flows through the ground. However, this will not fundamentally affect the benefits brought to both PEV owners and the DSO by our proposed PB scheme. Hence, we do not consider the multi-grounded network in this paper.⁵

Both electricity loads and REG are added to Bus 846, where the other loads in the original test feeder are kept unchanged. Note that when other buses are chosen, similar results can still be obtained. Due to the variability and uncertainty of weather conditions, customer behaviors, etc., inflexible loads on three phases of Bus 846 are assumed to be highly unbalanced. In our simulation, 100 sets of inflexible load data are generated randomly following Gaussian distributions to test our proposed scheme. The mean inflexible load on each phase is shown in Table I, while the standard deviation is 20 kW.⁶ For example, the inflexible load in time-slot 1 on phase A is generated following a distribution of $N(150, 20^2)$.

The maximum charging rate of PEVs is fixed at 50 kW. The battery capacity of each PEV is set as 100 kWh while the charging demands over 2 hours are generated following

⁵For a multi-grounded network, the technique proposed in [35] can be applied, where the grounding effect is taken into account by constructing a novel phase impedance matrix with modified Carson's equations.

⁶The setup in Table I is to make the inflexible loads unbalanced. The goal is to show the phase balancing capability of our proposed scheme. The particular mean value and standard deviation are chosen arbitrarily. Other kinds of values can achieve similar results.

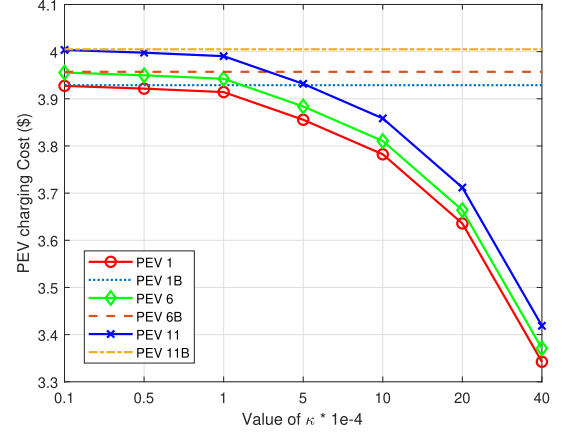


Fig. 3. Expected charging costs of PEV owners. The dotted lines, PEV 1B, PEV 6B and PEV 11B are the expected charging costs of PEV 1, PEV 6 and PEV 11 respectively in the benchmark case, where no PB remuneration term is included.

the Gaussian distribution $\mathcal{N}(40, 4^2)$. A total of $N = 15$ PEVs are assumed to be available and distributed evenly over three phases, thus we have $\mathcal{N}_1 = \{1, \dots, 5\}$, $\mathcal{N}_2 = \{6, \dots, 10\}$ and $\mathcal{N}_3 = \{11, \dots, 15\}$. In this simulation, we select a linear pricing function, i.e., $p_t(L_t) = aL_t + b$, where $a = 10^{-5}$ \$/kW²h, and $b = 0.1$ \$/kWh. Constants η , ϵ and δ in Algorithm 1 are set as 1.3 , 10^{-3} and 10^{-3} respectively.

A. Financial Effects of Phase Balancing

In this subsection, we consider the effect of phase balancing on the charging costs of PEV owners and the operating cost of the DSO. Note that the charging cost we used here and in later parts also includes the remuneration term, which is regarded as negative cost. In Fig. 3, we increase the value of PB pricing coefficient κ from 10^{-5} \$/kW²h to 4×10^{-3} \$/kW²h. Without loss of generality, three PEVs from three different phases, i.e., PEV 1, 6 and 11, are selected for illustration. The benchmark case is a non-cooperative game among PEV owners, where each owner minimizes the charging cost considering others' charging decisions, with no PB remuneration term included. We can see that the expected charging cost of each PEV owner over different sets of inflexible load data is less than that in the benchmark case. This is easy to understand, since the inclusion of PB remuneration provides PEV owners a new way to take advantage of the battery charging flexibility.

On the other hand, PB also brings financial benefits to the DSO. In fact, by performing a three-phase power flow on the modified 34-bus test feeder, Fig. 4 shows that with the increase of κ , the expected total energy loss in this distribution system decreases. Here the energy loss E_{loss} is defined as follows:

$$E_{\text{loss}} = \sum_{t \in \mathcal{T}} \sum_{l \in \mathcal{L}} I_{l,t}^2 R_l,$$

where $\mathcal{L}$ is the set of distribution lines, $I_{l,t}$ is the current magnitude in line l at time t , R_l is the resistance of line l . As a result, the expected energy cost of the DSO is reduced as well. In Fig. 5, the cost reduction of the DSO is defined as the

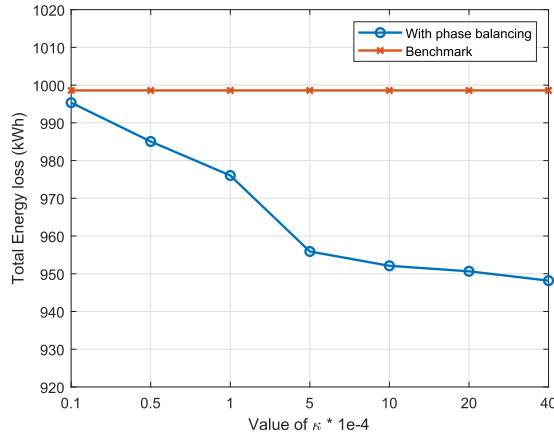


Fig. 4. Expected energy loss in the distribution system over the 2-hour time horizon when the value of κ varies.

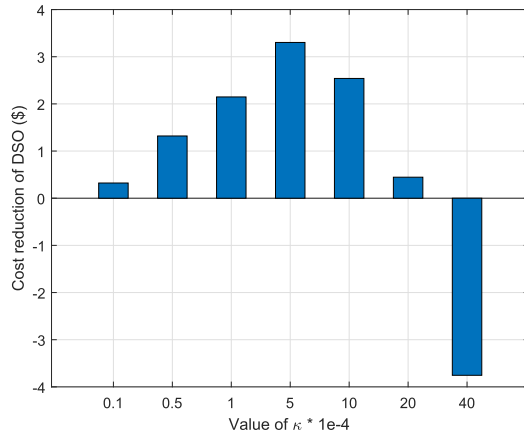


Fig. 5. The expected cost reduction of the DSO when the value of κ varies.

reduced energy cost minus the remuneration provided to the PEV owners. The energy price is set as 0.1\$/kWh. We can see that in most cases, the expected cost reduction of the DSO is positive, which means the monetary savings from reduced energy loss is more than enough to cover the PB incentives paid to the PEV owners. Therefore, the DSO can also benefit financially from our proposed PB scheme. Note that when the value of κ is too large, the loads are relatively balanced, the energy loss reduction capability is almost saturated, while the remuneration paid to the PEV owners becomes a large amount, hence the cost reduction will be negative.

B. Power Quality Effects of Phase Balancing

In this subsection, we investigate the effects of PB on the power quality of the distribution system.

We first consider the voltage imbalance (VIB), which is defined in (17) based on IEEE Std. 112-1991 [36]:

$$\text{VIB} = \frac{\text{Max}(|V_A - V_{\text{avg}}|, |V_B - V_{\text{avg}}|, |V_C - V_{\text{avg}}|)}{V_{\text{avg}}}, \quad (17)$$

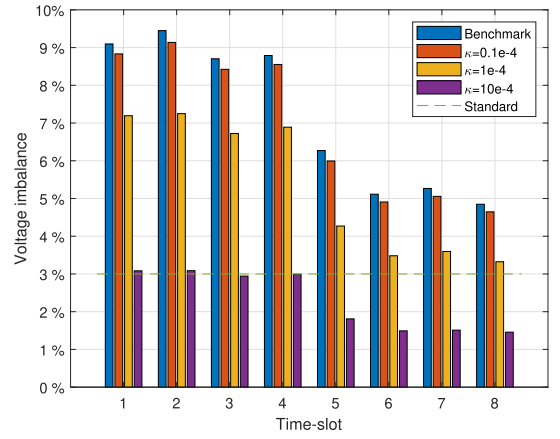


Fig. 6. Voltage imbalance over time at bus 846.

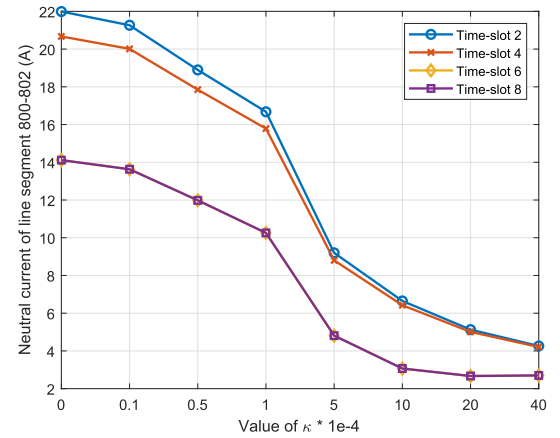


Fig. 7. Neutral currents in line segment 800-802 when the value of κ varies.

where V_A , V_B and V_C are the voltages on three different phases, and V_{avg} is their average. Fig. 6 illustrates the voltage imbalance at bus 846 over the time horizon. In this figure, the voltage imbalance decreases when κ increases. The reason is that when κ becomes larger, PEV owners are more willing to help in reducing the phase imbalance. This result follows easily from the definition of (7). As the maximum allowable VU of electric supply systems is 3% according to [37], we can conclude that our proposed scheme is able to reduce the voltage imbalance to the safe zone.

Then, we present the effect of our PB scheme on the neutral currents as shown in Fig. 7.⁷ The line segment 800-802 is taken as an example. In this figure, compared with the benchmark where no PB is introduced, i.e., the case when $\kappa = 0$, the neutral current decreases by a significant ratio when κ is large enough. Since the reduced neutral current leads to less probability of feeder tripping, our proposed scheme can improve the security and reliability of the distribution system.

⁷For multi-grounded networks, this figure alone is not sufficient as the effect of ground current must be considered. However, when the loads become more balanced, the sum of neutral current and ground current will be smaller, which conforms with the main idea of this figure.

TABLE II
MEAN INFLEXIBLE LOADS AT BUS 846 WHEN DIFFERENT PRICING FUNCTIONS ARE USED (UNIT: kW)

Time-slot	Phase A	Phase B	Phase C
1-4	150	50	250
5-8	100	150	50

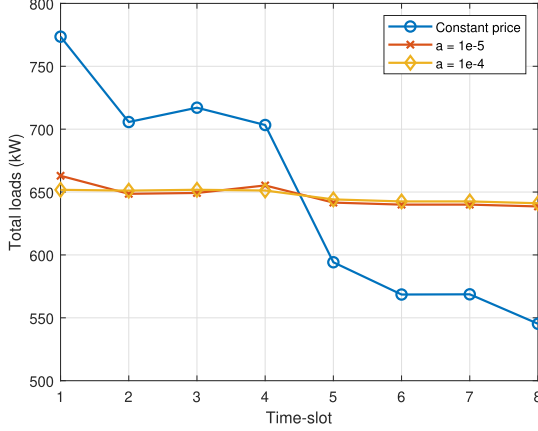


Fig. 8. Load profiles at Bus 846 with different pricing functions.

C. Effects of Different Pricing Functions

In this subsection, we illustrate how our proposed scheme performs under different energy pricing functions. Three different kinds of pricing functions $p_t(L_t) = aL_t + b$ are used in the simulation: the constant energy price function which does not change with power load, i.e., $a = 0$; and two inclining energy price functions which increase with the power load, that is, $a = 10^{-5}$ \$/kW²h and $a = 10^{-4}$ \$/kW²h. The inflexible loads at Bus 846 are generated with the mean value shown in Table II and the standard deviation being 20 kW.⁸ The parameter κ is fixed at 10^{-3} \$/kW²h. Other setups are similar with those in subsection A and B.

In Fig. 8, one case of total load profiles with three different pricing functions is presented. In this scenario, inflexible loads from time-slot 1 to 4 are higher than those from time-slot 5 to 8. When the energy price is constant, the energy cost becomes a linear function of the total consumed energy, and it is not affected by how the charging process is shifted. In this way, PEV owners only care about reducing the phase imbalance to acquire more PB remuneration. Therefore, it is possible that an increase in peak demand may appear, e.g., in time-slot 1 of Fig. 8. On the contrary, when the energy price is inclining, an increase in peak demand incurs more energy costs. As a result, PEV owners will try to avoid the high load peak during their charging scheduling. Following the same reasoning, from Fig. 8, we can also see that as a increases, the energy price inclines faster with the power load, then the load profile becomes more flat.

Similar with the discussions for Fig. 8, Fig. 9 can be explained easily. It compares the voltage imbalance when different pricing functions are used. For the constant price case, PEV owners only

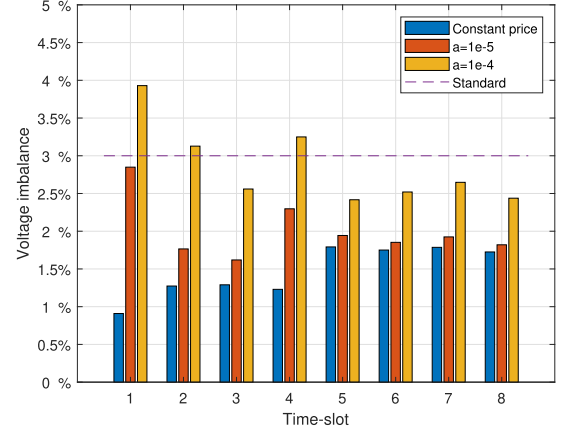


Fig. 9. Voltage imbalance over time at Bus 846 with different pricing functions.

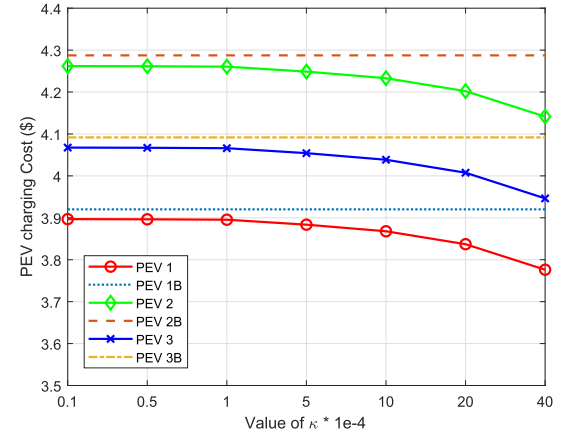


Fig. 10. Expected charging costs of PEV 1, 2, 3 at Bus 808 when loads are distributed.

care about PB, thus the voltage imbalance is the lowest. When a increases, the effect of peak load becomes more significant, so the voltage imbalance also increases.

In summary, to avoid the possible peak load increase, inclining pricing functions should be used. The parameter a controls the tradeoff of PEV owners between energy cost and PB remuneration, which is similar with the role of κ . In practice, when setting the value of κ , the effect of a should be fully considered.

D. Effects of Load Distribution Over the Network

In this subsection, we study the case where inflexible loads and PEV loads are more distributed over the network. Instead of adding all the loads to Bus 846, inflexible loads and 15 PEVs are divided equally in 5 parts, and added to Bus 808, 816, 854, 846 and 836, respectively. At each bus, the proposed PB scheme is performed independently among the three PEVs charged at that bus. After the power loads at each bus have been determined, we run a three-phase power flow program to obtain the system energy loss and calculate the corresponding DSO cost reduction.

As shown in Fig. 10, 11 and 12, we can obtained similar results with those in subsection A. To be specific, under our proposed

⁸The reason that Table II is different from Table I is to highlight the possibility of an increase in peak demand when the energy price is constant.

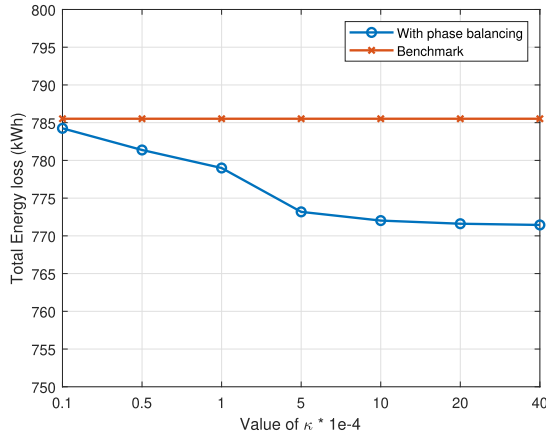


Fig. 11. Expected energy loss in the distribution system when loads are distributed.

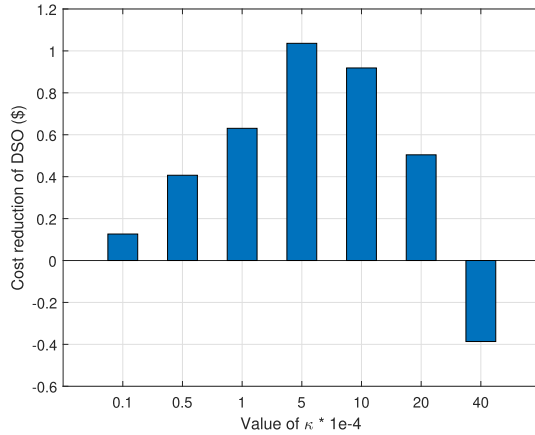


Fig. 12. Expected cost reduction of the DSO when loads are distributed.

PB scheme, the PEV charging costs are reduced, as well as the system energy loss and the DSO's operation cost. Since Bus 846 locates at the end of this distribution system, loads added there incur more energy loss than the same loads added in the front buses of the system, such as Bus 808 and Bus 816. Therefore, the energy loss here is less than that in subsection A, so the potential of PB for reducing energy loss is also smaller. As a result, the amount of charging cost reduction and DSO cost reduction from PB are both less than those illustrated in subsection A.

E. Implementation Under Load Uncertainty

In this subsection, we consider the implementation issues when load uncertainty is around. As discussed in Section III-D, in practice, we can use a receding horizon based strategy to tackle the load uncertainty. At each time-slot, loads of the current time-slot and mean loads of the future time-slots are assumed to be known. With this information, the PB game is solved, and only the results of the first time-slot are implemented. At the next time-slot, this process is repeated with a shifted time horizon. In Fig. 13, three different implementation strategies are performed: the receding horizon based strategy; the offline strategy where

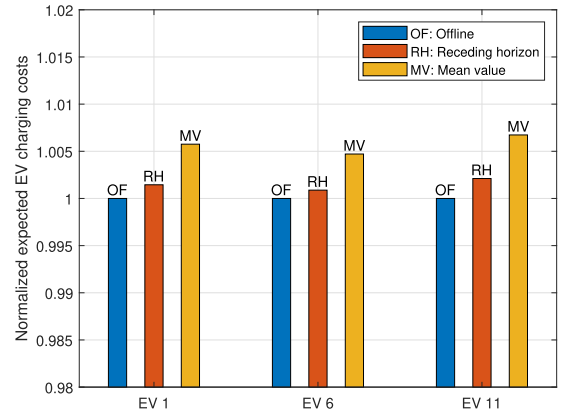


Fig. 13. Normalized expected PEV charging costs with different implementation strategies under load uncertainty.

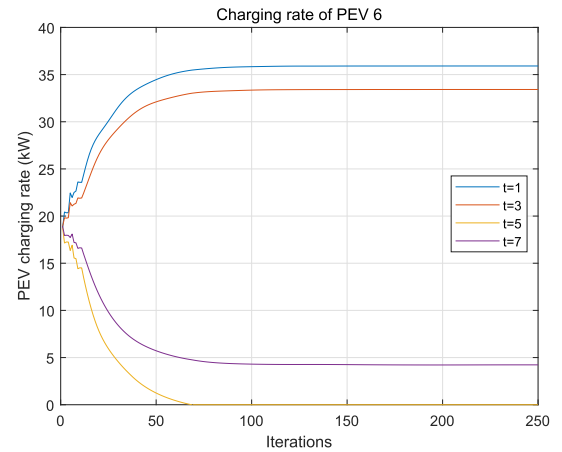


Fig. 14. Convergence of the charging rates of PEV 6 in time-slots 1, 3, 5, 7.

all the future information are assumed to be known; the mean value strategy where all the unknown parameters are replaced by their mean values. Other setups are the same with those in subsection A and B. Like in Fig. 3, PEV 1, 6 and 11 are selected as examples. Their expected charging costs are normalized with the offline costs set as 1. From this figure, we can find that the performance of the receding horizon based strategy is similar with the optimal offline one, and better than the mean value one. Hence, this result justifies its effectiveness to handle load uncertainty.

F. Convergence Results

In this subsection, we provide the convergence results of the proposed distributed algorithm. As shown in Fig. 14, without loss of generality, PEV 6 and time-slots 1, 3, 5, 7 are selected as an example. When other PEVs or other time-slots are chosen, similar results can still be obtained. In this figure, the algorithm converges in less than 100 iterations. Note that iteration here refers to the inner iteration in this bilevel algorithm. The average running time of each iteration is 0.0065 s, where the simulation is run by Matlab 2018a on a Laptop with a 2.3 GHz Intel Core i5 CPU and 8 GB Memory. Therefore, our proposed

algorithm is able to run in a receding horizon fashion, which ensures that dynamic phase balancing is achieved.

V. CONCLUSION

In this paper, we proposed an PEV-based phase balancing scheme. The introduction of PB remuneration incentivizes PEV owners to help the DSO reduce the phase imbalance autonomously. We formulated this problem as a non-cooperative PB game, and introduced an extended game to circumvent the difficulty in handling the non-smoothness of the original one. Subsequently, connections between the NE of these two games were built. Based on this result, we proved the existence of PB game's NE and developed a distributed algorithm to achieve it. Convergence proof of this algorithm was also provided. Simulation results validated our proposed scheme, and illustrated that both the charging costs of PEVs and the operating cost of the DSO are reduced. Furthermore, we demonstrated that the power quality of the distribution system can be improved as well.

APPENDIX A PROOF OF THEOREM 1

Let $\mathbf{x}^i \in \Omega_i$ and $\mathbf{y}_t \in \Delta_t$ be arbitrary. For PEV i , we have

$$\begin{aligned} & \sum_{t=1}^T \left(p_t(x_t^i, x_t^{*, -i})x_t^i + \frac{\kappa_t}{N} \max_{\mathbf{y}_t \in \Delta_t} \sum_{j \in \Phi} y_t^j g_t^j(x_t^i, \mathbf{x}_t^{*, -i}) \right) \\ & \geq \sum_{t=1}^T \left(p_t(x_t^i, x_t^{*, -i})x_t^i + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^{*, j} g_t^j(x_t^i, \mathbf{x}_t^{*, -i}) \right) \\ & \geq \sum_{t=1}^T \left(p_t(x_t^{*, i}, x_t^{*, -i})x_t^{*, i} + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^{*, j} g_t^j(x_t^{*, i}, \mathbf{x}_t^{*, -i}) \right) \\ & = \sum_{t=1}^T \left(p_t(x_t^{*, i}, x_t^{*, -i})x_t^{*, i} + \frac{\kappa_t}{N} \max_{\mathbf{y}_t \in \Delta_t} \sum_{j \in \Phi} y_t^j g_t^j(x_t^{*, i}, \mathbf{x}_t^{*, -i}) \right), \end{aligned}$$

where this result follows from the fact that $\mathbf{x}^{*, i}$ and $\mathbf{y}_t^*$ constitute an NE of $\hat{\Gamma}$, and it leads to the theorem easily.

APPENDIX B PROOF OF LEMMA 1

Without loss of generality, we first select the objective function of PEV 1, i.e., $H_1(\mathbf{x}^1, \mathbf{x}^{-1}, \mathbf{y})$ as an example, where PEV 1 belongs to the set $\mathcal{N}_1$. Let $H(\mathbf{x}^1) = H_1(\mathbf{x}^1, \mathbf{x}^{-1}, \mathbf{y})$ when $\mathbf{x}^{-1}$ and $\mathbf{y}$ are fixed, that is,

$$\begin{aligned} H(\mathbf{x}^1) &= \sum_{t=1}^T \\ & \times \left[p_t(x_t^1, \mathbf{x}_t^{-1})x_t^1 + \frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j \left(\tilde{L}_t^j + \sum_{i \in \mathcal{N}_j} x_t^i - \frac{L_t}{3} \right)^2 \right]. \end{aligned}$$

As $p_t(x_t^1, \mathbf{x}_t^{-1})$ is assumed to be continuously differentiable, $H(\mathbf{x}^1)$ is also continuously differentiable. It is easy to see that

$$\begin{aligned} \frac{d^2 H}{d(x_t^1)^2} &= p_t'' x_t^1 + 2p_t' + \frac{2\kappa_t}{9N} (4y_t^1 + y_t^2 + y_t^3) \geq 0, \\ \frac{d^2 H}{dx_t^1 dx_s^1} &= 0, \end{aligned}$$

where $s \neq t, s \in \mathcal{T}$. Therefore, the Hessian of $H(\mathbf{x}^1)$ is positive semi-definite, and the convexity of $H(\mathbf{x}^1)$ follows from [24]. For objective functions of other PEVs, we can demonstrate their convexity and continuous differentiability in similar ways.

As for the objective function of AP $t, t \in \mathcal{T}$, when $\mathbf{y}_{-t}$ and $\mathbf{x}$ are fixed, the function

$$I(\mathbf{y}_t) = -\frac{\kappa_t}{N} \sum_{j \in \Phi} y_t^j \left(\tilde{L}_t^j + \sum_{i \in \mathcal{N}_j} x_t^i - \frac{L_t}{3} \right)^2$$

is a linear function of $\mathbf{y}_t$. As a result, $I(\mathbf{y}_t)$ is also convex and continuously differentiable.

Based on the analyses above, Lemma 1 has been proved.

APPENDIX C PROOF OF LEMMA 2

According to Proposition 12.3 in [38], the mapping $\mathbf{F}(\mathbf{z})$ is monotone if and only if the Jacobian matrix $\mathbf{J}\mathbf{F}(\mathbf{z})$ satisfies that $\frac{1}{2}(\mathbf{J}\mathbf{F}(\mathbf{z}) + \mathbf{J}\mathbf{F}^T(\mathbf{z}))$ is positive semi-definite.

Given the definition of $\mathbf{F}(\mathbf{z})$, $\mathbf{J}\mathbf{F}(\mathbf{z})$ can be written as

$$\mathbf{J}\mathbf{F}(\mathbf{z}) = \begin{bmatrix} \mathbf{J}\mathbf{H}_{xx} & \mathbf{J}\mathbf{H}_{xy} \\ \mathbf{J}\mathbf{I}_{yx} & \mathbf{J}\mathbf{I}_{yy} \end{bmatrix}.$$

In this formulation, $\mathbf{J}\mathbf{H}_{xx} = [\mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} H_i]_{i,j \in \mathcal{N}}$, where

$$\mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} H_i = \left[\frac{\partial^2 H_i}{\partial x_s^i \partial x_t^j} \right]_{s,t \in \mathcal{T}} \in \mathbb{R}^{T \times T},$$

other submatrices can be written in similar ways.

It is easy to see that $\mathbf{J}\mathbf{I}_{yy} = \mathbf{0}$, and $\mathbf{J}\mathbf{H}_{xy} = -\mathbf{J}\mathbf{I}_{yx}^T$, hence

$$\frac{1}{2}(\mathbf{J}\mathbf{F} + \mathbf{J}\mathbf{F}^T) = \begin{bmatrix} \mathbf{J}\mathbf{H}_{xx} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix},$$

$\frac{1}{2}(\mathbf{J}\mathbf{F} + \mathbf{J}\mathbf{F}^T)$ is positive semi-definite if and only if $\mathbf{J}\mathbf{H}_{xx}$ is positive semi-definite. Let $f_i = \sum_{t=1}^T p_t x_t^i$, and $g_i = \sum_{t=1}^T \sum_{j \in \Phi} \kappa_t y_t^j g_t^j / N$, then $H_i = f_i + g_i$, $\mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} H_i = \mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} f_i + \mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} g_i$. Denote the matrices $[\mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} f_i]_{i,j \in \mathcal{N}}$ and $[\mathbf{J}_{\mathbf{x}^j} \nabla_{\mathbf{x}^i} g_i]_{i,j \in \mathcal{N}}$ as $\mathbf{A}_f$ and $\mathbf{A}_g$ respectively, thus $\mathbf{J}\mathbf{H}_{xx} = \mathbf{A}_f + \mathbf{A}_g$.

At first, we consider the matrix $\mathbf{A}_f$. From the definition of function f_i , $\mathbf{A}_f$ can be written as

$$\begin{aligned} \mathbf{A}_f &= (\mathbf{J}_N + \mathbf{I}_N) \otimes \text{diag}\{p_t'\}_{t \in \mathcal{T}} \\ & \quad + \mathbf{J}_{1,N} \otimes [\text{diag}\{p_t'' x_t^1\}_{t \in \mathcal{T}}, \dots, \text{diag}\{p_t'' x_t^N\}_{t \in \mathcal{T}}]^T, \end{aligned}$$

where $\otimes$ denotes the Kronecker product, $\mathbf{J}_N \in \mathbb{R}^{N \times N}$ and $\mathbf{J}_{1,N} \in \mathbb{R}^{1 \times N}$ are all-ones matrices, and $\mathbf{I}_N \in \mathbb{R}^{N \times N}$

is an identity matrix. For an arbitrary vector $\mathbf{u} = (u_1^1, \dots, u_T^1, \dots, u_1^N, \dots, u_T^N) \in \mathbb{R}^{NT}$, we have

$$\begin{aligned} \mathbf{u}^T \mathbf{A}_f \mathbf{u} &= \sum_{t=1}^T \sum_{n=1}^N p'_t (u_t^n)^2 \\ &\quad + \sum_{t=1}^T \sum_{n=1}^N (p'_t + p''_t x_t^n) \left(u_t^n \sum_{m=1}^N u_t^m \right). \end{aligned}$$

Define two sets as follows:

$$\mathcal{N}_t^+ = \left\{ n \in \mathcal{N} : u_t^n \sum_{m=1}^N u_t^m \geq 0 \right\}, \quad \mathcal{N}_t^- = \mathcal{N} \setminus \mathcal{N}_t^+,$$

then for different n , the following inequalities hold:

$$\begin{aligned} (p'_t + p''_t x_t^n) \left(u_t^n \sum_{m=1}^N u_t^m \right) &\geq p'_t u_t^n \sum_{m=1}^N u_t^m, \quad \forall n \in \mathcal{N}_t^+, \\ (p'_t + p''_t x_t^n) \left(u_t^n \sum_{m=1}^N u_t^m \right) &\geq -(p'_t + p''_t x_{t,\max}^n) \left| u_t^n \sum_{m=1}^N u_t^m \right|, \\ &\quad \forall n \in \mathcal{N}_t^-. \end{aligned}$$

The above inequalities lead to

$$\begin{aligned} \sum_{n=1}^N (p'_t + p''_t x_t^n) \left(u_t^n \sum_{m=1}^N u_t^m \right) &\geq \sum_{n \in \mathcal{N}_t^+} p'_t u_t^n \sum_{m=1}^N u_t^m \\ &\quad - \sum_{n \in \mathcal{N}_t^-} (p'_t + p''_t x_{t,\max}^n) \left| u_t^n \sum_{m=1}^N u_t^m \right|. \end{aligned} \quad (18)$$

If $\sum_{m=1}^N u_t^m \geq 0$, then for $n \in \mathcal{N}_t^+$, $u_t^n \geq 0$, for $n \in \mathcal{N}_t^-$, $u_t^n < 0$. Thus, $p'_t (\sum_{n \in \mathcal{N}_t^+} u_t^n \sum_{m=1}^N u_t^m - \sum_{n \in \mathcal{N}_t^-} |u_t^n \sum_{m=1}^N u_t^m|) \geq 0$. Note that:

$$\begin{aligned} \sum_{n \in \mathcal{N}_t^-} \left| u_t^n \sum_{m=1}^N u_t^m \right| &\geq - \sum_{n \in \mathcal{N}_t^+} u_t^n \sum_{m=1}^N u_t^m \geq - \left(\sum_{n \in \mathcal{N}_t^+} u_t^n \right)^2 \\ &\geq \sum_{n \in \mathcal{N}_t^+} (u_t^n)^2 |\mathcal{N}_t^+| \geq -N \sum_{n=1}^N (u_t^n)^2 \end{aligned}$$

where $|\mathcal{N}_t^+|$ is the cardinality of set $\mathcal{N}_t^+$. The first inequality follows from $\sum_{n=1}^N u_t^n = \sum_{n \in \mathcal{N}_t^+} u_t^n - \sum_{n \in \mathcal{N}_t^-} |u_t^n| \geq 0$, the second one is derived from $\sum_{n=1}^N u_t^n \leq \sum_{n \in \mathcal{N}_t^+} u_t^n$, and the third one comes from the Cauchy-Schwarz Inequality. Thus, the right-hand-side (RHS) of inequality (18) satisfies:

$$RHS \geq -X_{\max} p''_t N \sum_{n=1}^N (u_t^n)^2.$$

If $\sum_{m=1}^N u_t^m < 0$, we can obtain the same result. Therefore, the following inequality holds:

$$\mathbf{u}^T \mathbf{A}_f \mathbf{u} \geq \sum_{t=1}^T (p'_t - N X_{\max} p''_t) \sum_{n=1}^N (u_t^n)^2. \quad (19)$$

Apparently, when $p'_t \geq N X_{\max} p''_t$ holds, $\mathbf{u}^T \mathbf{A}_f \mathbf{u} \geq 0$ for $\forall \mathbf{u}$. In other words, the matrix $\mathbf{A}_f$ is positive semi-definite.

Next we will consider another part of $\mathbf{JH}_{xx}$, i.e., $\mathbf{A}_g$. Let $g_{i,t} = \sum_{j \in \Phi} \kappa_t y_t^j g_t^j / N$, then $g_i = \sum_{t=1}^T g_{i,t}$. The matrix $\mathbf{A}_g$ can be written as $\mathbf{A}_g = \sum_{t=1}^T \mathbf{JG}_t \otimes \mathbf{J}_{1t}$, where $\mathbf{J}_{1t} \in \mathbb{R}^{T \times T}$ is a matrix with only one non-zero element which is $\mathbf{J}_{1t}(t, t) = 1$, $\mathbf{JG}_t \in \mathbb{R}^{N \times N}$ is defined as follows:

$$\mathbf{JG}_t = \left[\frac{\partial^2 g_{i,t}}{\partial x_t^j \partial x_t^k} \right]_{j,k \in \mathcal{N}} = \frac{2\kappa_t}{9N} \sum_{l \in \Phi} y_t^l \mathbf{JG}_t^l,$$

where the three matrices $\mathbf{JG}_t^l, l \in \Phi = \{1, 2, 3\}$ are:

$$\begin{aligned} \mathbf{JG}_t^1 &= \begin{bmatrix} 4\mathbf{J}_{|\mathcal{N}_1|} & -2\mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_2|} & -2\mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_3|} \\ -2\mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_1|} & \mathbf{J}_{|\mathcal{N}_2|} & \mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_3|} \\ -2\mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_1|} & \mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_2|} & \mathbf{J}_{|\mathcal{N}_3|} \end{bmatrix}, \\ \mathbf{JG}_t^2 &= \begin{bmatrix} \mathbf{J}_{|\mathcal{N}_1|} & -2\mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_2|} & \mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_3|} \\ -2\mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_1|} & 4\mathbf{J}_{|\mathcal{N}_2|} & -2\mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_3|} \\ \mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_1|} & -2\mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_2|} & \mathbf{J}_{|\mathcal{N}_3|} \end{bmatrix}, \\ \mathbf{JG}_t^3 &= \begin{bmatrix} \mathbf{J}_{|\mathcal{N}_1|} & \mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_2|} & -2\mathbf{J}_{|\mathcal{N}_1|,|\mathcal{N}_3|} \\ \mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_1|} & \mathbf{J}_{|\mathcal{N}_2|} & -2\mathbf{J}_{|\mathcal{N}_2|,|\mathcal{N}_3|} \\ -2\mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_1|} & -2\mathbf{J}_{|\mathcal{N}_3|,|\mathcal{N}_2|} & 4\mathbf{J}_{|\mathcal{N}_3|} \end{bmatrix}. \end{aligned}$$

Note that $|\mathcal{N}_l|$ is the cardinality of the set $\mathcal{N}_l$, while $\mathbf{J}_{|\mathcal{N}_l|} \in \mathbb{R}^{|\mathcal{N}_l| \times |\mathcal{N}_l|}$ and $\mathbf{J}_{|\mathcal{N}_k|,|\mathcal{N}_l|} \in \mathbb{R}^{|\mathcal{N}_k| \times |\mathcal{N}_l|}$ are all-one matrices.

Obviously the following results hold:

$$\text{Rank}(\mathbf{JG}_t^l) = 1, \quad \forall l \in \Phi, \quad (20)$$

$$\text{Trace}(\mathbf{JG}_t^l) = N + 3|\mathcal{N}_l| > 0, \quad \forall l \in \Phi. \quad (21)$$

Hence, matrix $\mathbf{JG}_t^l$ has only one non-zero eigenvalue which is positive. This fact demonstrates that $\mathbf{JG}_t^l$ is positive semi-definite, so is the matrix $\mathbf{JG}_t$ since every y_t^l is nonnegative. Finally, the positive semi-definiteness of $\mathbf{A}_g$ also follows.

In summary, when $p'_t \geq N X_{\max} p''_t$ holds, both $\mathbf{A}_f$ and $\mathbf{A}_g$ are positive semi-definite. Therefore, the mapping $\mathbf{F}_z$ is monotone which concludes the proof.

APPENDIX D PROOF OF THEOREM 2

Following Proposition 7 and Theorem 17 in [29] and Lemma 1, when $p'_t \geq N X_{\max} p''_t$ and $0 < \eta < 2$ hold, Algorithm 1 converges to an NE of game Γ if

$$\sum_{m \in (\mathcal{N} \cup \mathcal{T}) \setminus \{n\}} \frac{\beta_{nm}^{\max}}{\tau + \alpha_n^{\min}} < 1, \quad \forall n \in \mathcal{N} \cup \mathcal{T}, \quad (22)$$

where $\beta_{nm}^{\max}$ and $\alpha_n^{\min}$ are defined as follows:

$$\beta_{nm}^{\max} = \max_{\mathbf{z} \in \mathbf{K}} \|\mathbf{J}_m \mathbf{F}_n\|, \quad \alpha_n^{\min} = \min_{\mathbf{z} \in \mathbf{K}} \lambda_{\min}(\mathbf{J}_n \mathbf{F}_n).$$

In these two definitions, $\|\cdot\|$ is the Euclidean norm, $\lambda_{\min}(\cdot)$ denotes the smallest eigenvalue, and

$$\mathbf{J}_m \mathbf{F}_n = \begin{cases} \mathbf{J}_m \nabla_{\mathbf{x}^n} \mathbf{H}_n, & \text{if } n \in \mathcal{N}, m \in \mathcal{N} \cup \mathcal{T}, \\ \mathbf{J}_m \nabla_{\mathbf{y}^n} \mathbf{I}_n, & \text{if } n \in \mathcal{T}, m \in \mathcal{N} \cup \mathcal{T}. \end{cases}$$

From Appendix B, we can easily conclude that $\mathbf{F}_n(\mathbf{x}_n)$ is monotone with respect to $\mathbf{x}_n$, thus $\alpha_n^{\min} \geq 0$.

For $\beta_{nm}^{\max}$, $n \in \mathcal{N}$, $m \in \mathcal{N}$, we have

$$\beta_{nm}^{\max} \leq \max_{t \in \mathcal{T}} \max_{\mathbf{z} \in \mathbf{K}} \sqrt{N} \left(p'_t + p''_t x_t^n + \sum_{l=1}^3 \frac{y_t^l \kappa_t}{N} \left| \frac{\partial^2 g_t^l}{\partial x_t^n \partial x_t^m} \right| \right) \\ \leq \sqrt{N} (P_1 + P_2 X_{\max} + \kappa_M / N),$$

where the first inequality follows from the fact that $\mathbf{J}_m \mathbf{F}_n$ is a diagonal matrix, and the last inequality follows from $|\partial^2 g_t^l / (\partial x_t^n \partial x_t^m)| < 1$ and $\sum_{l=1}^3 y_t^l = 1$.

Similarly, for $\beta_{nm}^{\max}$, $n \in \mathcal{N}$, $m \in \mathcal{T}$, we have

$$\beta_{nm}^{\max} \leq \frac{4\kappa_M}{3N} \sqrt{\frac{2}{3}} (NX_{\max} + \tilde{L}_{\max}).$$

For $\beta_{nm}^{\max}$, $n \in \mathcal{T}$, $m \in \mathcal{N}$, we have $\beta_{nm}^{\max} = \beta_{mn}^{\max}$. Finally, for $\beta_{nm}^{\max}$, $n, m \in \mathcal{T}$, we have $\beta_{nm}^{\max} = 0$. Therefore the left-hand-side (LHS) of inequality (22) becomes:

$$LHS \leq \frac{1}{\tau} \left(N^{3/2} (P_1 + P_2 X_{\max}) + N\kappa_M \right. \\ \left. + \frac{4T\kappa_M}{3N} \sqrt{\frac{2}{3}} (NX_{\max} + \tilde{L}_{\max}) \right), \forall n \in \mathcal{N} \cup \mathcal{T}.$$

Apparently, when the convergence condition in Theorem 2 holds, $LHS < 1, \forall n \in \mathcal{N} \cup \mathcal{T}$. Hence, Algorithm 1 converges to an NE of game Γ .

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